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Unaccusativity in Japanese: Evidence from L2 Grammar*

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1. Introduction

Argument structure, which outlines the configuration of arguments associated with verbs, has been one of the central topics in syntax, semantics, and L2 acquisition. Among others, two kinds of intransitive verbs—*unergative verbs* and *unaccusative verbs*—have been explored in depth. Intransitive verbs select a single argument, but the syntactic position of the arguments varies depending on the θ-roles they receive. At the level of underlying structures, unergative verbs require an external argument in Spec-vP (1a), whereas unaccusative verbs select an internal argument in Comp-VP (1b, Burzio, 1986; Perlmutter, 1978):

(1) a. [$_{\text{vP}}$ a boy [$_{\text{v}'}$ ran ___]] *unergative*
  b. [$_{\text{vP}}$ ____[$_{\text{v}'}$ arrive a train]] *unaccusative*

However, the difference in underlying syntactic composition is lost on the surface in English due to the presence of obligatory subject movement:

(2) a. [$_{\text{TP}}$ a boy [$_{\text{vP}}$ ____ [$_{\text{v}'}$ ran]]] *unergative*
  b. [$_{\text{TP}}$ a train [$_{\text{vP}}$ arrive [$_{\text{v}'}$ ____]] *unaccusative*

A similar issue occurs in Japanese. In Japanese, subjects may remain in situ (see e.g., Fukui, 1995; Yatsushiro, 1999). Moreover, Japanese is a head-final language, and subject movement is missing. As a consequence, the distinction between unergative and unaccusative vP constructions is obscured in surface word orders; NP$_{\text{AGENT}}$ in Spec-VP precedes V in unergatives (3a), while V is linearized to the right of NP$_{\text{THEME}}$ in unaccusatives (3b). In both cases, NPs precede Vs.

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(3) a. [$_{vP}$ Syonen-ga warau ____]. *unergative*
boy-NOM laugh
'A boy laughs.'
b. [$_{vP}$ ____ mado-ga wareta]. *unaccusative*
window-NOM broke
'A window broke.'

Thus, unaccusativity is concealed in these languages.

As will be detailed below, the evidence for unaccusativity in Japanese is considered weak (Takami & Kuno, 2017). Against this background, this study aims to offer new evidence for unaccusativity in Japanese by examining Japanese–English interlanguage grammar. The following sections are organized as follows: Section 2 overviews the previous evidence for unaccusativity in Japanese and counterexamples against it; Section 3 briefly reviews previous L2 studies; Section 4 describes the experiment; and Section 5 concludes the paper.

2. Evidence for Unaccusativity in Japanese

Syntacticians have presented evidence for deep unaccusativity in Japanese. For instance, Kishimoto (1996) observed that *-kake* 'be about to' in '*V*-kake-no-*N*' constructions can attach to unaccusatives, but not to unergatives.

(4) a. *kare-kake-no* hana (unaccusative)
wither-KAKE-GEN flower
'a flower, almost withered'
b. **sakebi-kake-no* kankyaku (unergative)
shout-KAKE-GEN spectator
'spectators, who are about to shout'
(Kishimoto, 1996: 257)

However, Takami and Kuno (2017) argued against this observation, offering several counterexamples. First, they pointed out that unergatives (e.g., *stand up*, *sit*) can also appear in the *V*-kake-no-*N* construction:

(5) a. *Tati-kake-no* kirin-o syasin-ni totta yo.
stand.up-KAKE-GEN giraffe-ACC photo-DAT took SFP
'I took a picture of a giraffe in the process of standing up.'
b. Isutori-geemu-de *suwari-kake-no* hito-no isu-o totta.
musical.chairs-in sit-KAKE-GEN person-GEN chair-ACC pulled.away
'I pulled away a chair from the person who was about to sit on it in the game of musical chairs.'
(Takami & Kuno, 2017: 206–7)

Furthermore, they offered examples showing that even unaccusatives sometimes fail to allow the *V*-kake-no-*N* construction.

(6) *oti-kake-no enpitu
fall-KAKE-GEN pencil
a pencil in the middle of falling' (*ibid*: 208)

These examples suggest that the (un)grammaticality of the *V*-kake-no-*N* construction cannot serve as strong evidence for the unergative-unaccusative distinction.

Let us now consider another piece of evidence: quantifier constructions. A quantifier like *ippai* or *takusan-no* 'a lot' modifies NPs or VPs, but what it modifies differs in unaccusativity (Kageyama, 1996; Hirakawa, 2001; Kishimoto, 2003, 2005). In unergative constructions, it modifies the VP (7a), whereas in unaccusative constructions, it modifies the subject NP (7b).

(7) a. pro *ippai* asonda. (unergative – modifies the quantity of playing)
a.lot played
'(We) played a lot.'
b. pro *ippai* kusatta. (unaccusative – modifies the number of the subject)
a.lot rotted
'A lot of things rotted.'

The interpretive difference has been attributed to unaccusativity: *ippai* can modify the NP within VP (internal argument) in unaccusatives (7b), whereas it fails to in unergatives (7a), where the NP is an external argument outside of VP.

In contrast, Takami and Kuno (2017: 207) offered the following counterexamples to this generalization, showing that the opposite interpretation is permitted. In (8a), which contains an unergative verb, the quantifier modifies the number of the subject, and in (8b), which involves an unaccusative verb, it modifies the quantity of VP:

(8) a. Petto-ga *ippai* nigeta. (unergative – modifies the number of the subject)
pet-NOM a.lot ran.away
'A lot of pets ran away.'
b. pro *ippai* naita. (unaccusative – modifies the quantity of crying)
a.lot cried
'(I) cried a lot.'

Thus, the *ippai/takusan-no* test does not appear to provide valid evidence for unaccusativity in Japanese.

In summary, the evidence presented thus far does not strongly support unaccusativity in Japanese. Against this backdrop, this study explores the possibility that Japanese-English interlanguage provides indirect supporting evidence.

3. Considering Japanese-English Interlanguage

Recall that deep unaccusativity is obscured by obligatory subject movement in English. Obligatory subject movement in English is triggered by uninterpretable φ-features (*u*φ), which are extremely difficult for Japanese-speaking learners of English (JLEs) at the intermediate level of proficiency to acquire. This is consistent with established findings that the acquisition of a new uF in L2 poses significant challenges for learners (see, e.g., Hawkins & Chan, 1997; Hawkins & Hattori, 2006; Tsimpli & Dimitrakopoulou, 2007; Al-Thubaiti, 2020; Kimura, 2022a, b, *to appear*; Kimura & Wakabayashi, 2024, *to appear*). If JLEs fail to acquire this feature, they may leave the subject in its original position (9a), or they may (optionally) manage to front the subject to Spec-TP via a non-target-like operation such as scrambling (9b).

(9) a. [$_{TP}$ [$_{VP}$ arrived a train]]
    b. [$_{TP}$ a train [$_{VP}$ arrived ___]]

Elementary JLEs are observed to employ optional scrambling to dislocate *wh*-phrases (Miyamoto & Iijima, 2003)—sometimes leaving them in situ and sometimes fronting them—in English *wh*-questions. By the same logic, it is likely that elementary JLEs fail to acquire φ-features and instead employ optional scrambling for subject movement.

Recall also that unaccusativity is obscured by head-finality in Japanese. In head-final languages, by linearizing V to the left of other elements, the verb appears in the final position in both unergative and unaccusative constructions. However, this issue seems to be resolved in the Japanese-English interlanguage. It is well known that L2 learners reset the headedness parameter at the earliest stages of L2 acquisition (Haznedar, 1997). That is, even elementary JLEs are expected to have reset headedness. If this assumption holds, and subject movement is missing, the *[$_{VP}$ NP [$_{V'}$ V]]* structure is linearized as it stands, and the *[$_{VP}$ [$_{V'}$ V NP]]* structure is also linearized as it stands, resulting in distinct word orders.

If our argument is correct, the Japanese-English interlanguage may reveal underlying syntactic structures regarding intransitive verbs. As illustrated in Table 1 below, while unaccusativity does not surface in Japanese or English due to headedness or obligatory subject movement, the Japanese-English interlanguage may exhibit it on the surface. This would be due to learners resetting headedness while struggling to acquire obligatory subject movement.

Table 1
Summary of relevant properties in Japanese, English and Japanese-English interlanguage

	headedness	subj-mvt	surface unaccusativity
Japanese	last	*	*
English	first	✓	*
J-E interlanguage	first?	*?	✓?

note. Shaded cells indicate factors obscuring unaccusativity.

A couple of notes are in order here. The subject-less configurations in (9a, b) may be rejected by learners for independent reasons: JLEs rarely drop subjects in L2 English (e.g., Wakabayashi & Negishi, 2003). To address this issue, let us consider Kuribara's (2004) finding that JLEs accepted configurations where non-subject phrases, such as adverbial topics, occupy the pre-verbal position (e.g., *Last Christmas had a wonderful party*). Taking these previous findings into account, we should modify the configurations in (9) to include an overt phrase in the pre-verbal position, as shown below:

(10) a. *[**When** did [$_{TP}$____[$_{VP}$ arrive the train]]]? *without scrambling/mvt*
b. [**When** did [$_{TP}$ the train [$_{VP}$ arrive____]]]? *with scrambling/mvt*

Let us now consider the unergative-unaccusative distinction. As noted earlier, English requires an NP to occupy the subject position (i.e., Spec-TP), regardless of its base position:

(11) a. [When did [$_{TP}$ the train [$_{VP}$ arrive ____]]]? *unaccusative*
b. [Where did [$_{TP}$ the man [$_{VP}$ ____ dance]]]? *unergative*

A key prediction is that elementary JLEs will leave the subject NP in situ.

If elementary JLEs are sensitive to the unergative-unaccusative distinction, they may accept a configuration like (12a), which reflects the underlying unaccusativity despite being ungrammatical due to the lack of subject movement.[1] In contrast, they will likely reject a configuration like (12b), which is incompatible with the underlying unergative structure.

(12) a. *[When did [$_{TP}$ ___ [$_{VP}$ arrive *the train*]]]? *unaccusative*
b. *[Where did [$_{TP}$ ___ [$_{VP}$ dance *the man*]]]? *unergative*

The unergative-unaccusative distinction may stem from learners' L1 Japanese or from principles of Universal Grammar (UG), including θ-theory (e.g., Baker,

[1] The lack of subject movement appears to stem from the failure to acquire the *u*φ feature, and the ungrammaticality does not result from a violation of a principle of UG.

1988; Progovac, 2015; Chomsky, 2021). Against this background, this study aims to examine whether interlanguage at the early stages of development exhibits sensitivity to the unergative-unaccusative distinction.

4. Experiment

4.1 Participants

Thirty-one native speakers of English (NSEs) and fifty JLEs participated in this study. The NSEs, residing in the U.S. at the time of testing, were recruited via Prolific. The JLEs, who were university students in Japan, were recruited from the authors' English classes and had no background in linguistics or related subjects. Both groups completed the task on the same platform (Google Forms).

The JLEs' English proficiency was assessed using the Oxford Quick Placement Test (University of Cambridge Local Examinations Syndicate, 2001). Based on the test criteria, 26 were classified as elementary-level learners, and 24 as intermediate-level learners.

4.2 Task and Materials

We conducted an untimed acceptability judgment task on a 5-point scale (1: Completely Unacceptable, 5: Completely Acceptable). The main test items included either unaccusative verbs (*arrive, happen, die, fall, melt, freeze, close, appear*) or unergative verbs (*swim, run, walk, dance, sing, cry, laugh, shout*; cf. Levin and Rappaport Hovav, 1995).[2] Based on the discussion above, we placed an adverbial *wh*-phrase (i.e., *when* or *where*) at the sentence-initial position.

Grammatical sentences contained the *Wh-Aux-DP-V* order (13a, c), whereas ungrammatical sentences contained the **Wh-Aux-V-DP* order (13b, d):

(13)	a.	Where did the train arrive?	*unaccusative/grammatical*
	b.	* When did happen the accident?	*unaccusative/ungrammatical*
	c.	Where did the man walk?	*unergative/grammatical*
	d.	* When did run the man?	*unergative/ungrammatical*

The number of *when* and *where* items was balanced across types and conditions. Eight sets of main test stimuli were created for each verb type. These were divided into two stimulus lists, with each type ultimately containing four tokens per list (16 in total). Thirty-six fillers, used in both lists, were also included, resulting in a total of 52 stimuli per participant. Test items were manually pseudo-randomized (i.e., within and between blocks; Ionin, 2012).

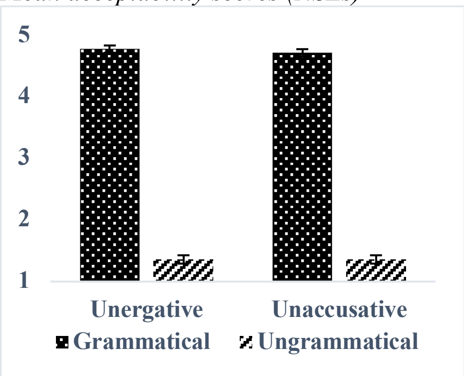
[2] As pointed out by the audience at BUCLD, we should have restricted the unaccusative verbs to non-alternating ones. We leave this issue for future research.

5. Results

Response data were analyzed using linear mixed-effects models in R (Baayen et al., 2008). Prior to the analysis, fixed effects (verb type and grammaticality) were dummy-coded: unaccusative verbs were coded as ‘0’ and unergative verbs as ‘1,’ while ungrammatical sentences were coded as ‘0’ and grammatical sentences as ‘1.’ The coded values were centered to estimate main effects. The maximum model included verb type, grammaticality, and their interaction terms as fixed effects, along with random slopes and intercepts for participants and items. The model was specified as follows: Response ~ Type * Grammaticality + (1 + Type * Grammaticality | Participants) + (1 + Type * Grammaticality | Item). The random effect structure for each group was selected using a *step* function. All models were implemented in R with the *lme4* package (Bates et al., 2020), and *p*-values were obtained using the *lmerTest* package (Kuznetsova et al., 2017).

NSEs displayed the expected response patterns, distinguishing grammatical from ungrammatical sentences (Figure 1). The statistical estimates from the final model (Response ~ Type * Grammaticality + (1 + Grammaticality | Participants)) indicate that only the main effect of grammaticality was significant ($\beta = 3.39$, $t = 37.98$, $p < 0.001$). Neither the main effect of verb type ($\beta = 0.03$, $t = 0.56$, $p = 0.58$) nor the interaction effect ($\beta = 0.07$, $t = 0.62$, $p = 0.54$) reached significance, as expected.

Figure 1
Mean acceptability scores (NSEs)



note. Error bars stand for standard errors of the mean.

Turning to the L2 results, elementary JLEs exhibited markedly different behaviors from NSEs (Figure 2), whereas intermediate JLEs displayed a pattern similar to that of NSEs (Figure 3). Notably, elementary JLEs made a clear distinction between ungrammatical unergative sentences and the other types. The results of the statistical estimation (final model: Response ~ Type * Grammaticality + (1 + Grammaticality | Participants)) revealed significant main effects for verb type ($\beta = -0.31$, $t = -2.62$, $p < 0.01$) and grammaticality ($\beta = 0.62$, $t = 2.48$, $p = 0.02$), as well as their interaction ($\beta = 0.85$, $t = 3.60$, $p < 0.001$).

The results of pairwise comparisons (final model: Response ~ Pairwise_Type + (Grammaticality | Participants)) showed that the difference in responses between grammatical and ungrammatical unaccusatives did not reach significance ($\beta = 0.22$, $t = 0.79$, $p = 0.44$). However, the difference between ungrammatical unaccusatives and unergatives was significant ($\beta = -0.72$, $t = -4.36$, $p < 0.01$).

These findings suggest that elementary JLEs permitted both grammatical and ungrammatical configurations for unaccusatives but only accepted grammatical configurations for unergatives.

Figure 2
Mean acceptability scores (elementary JLE)

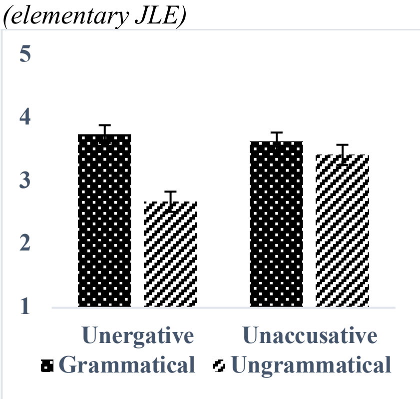
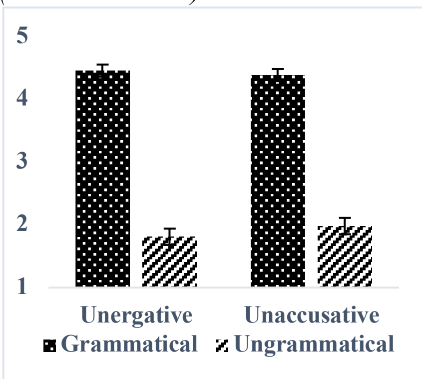


Figure 3
Mean acceptability scores (intermediate JLE)



Next, results from intermediate JLEs (final model: Response ~ Type * Grammaticality + (Type + Grammaticality | Participants)) showed that the main effect of grammaticality was significant ($\beta = 2.53$, $t = 10.2$, $p < 0.001$), whereas the main effect of verb type and the interaction were not significant ($\beta = -0.05$, $t = -0.39$, $p = 0.70$ for type; $\beta = 0.24$, $t = 1.2$, $p = 0.23$ for the interaction). This suggests that intermediate JLEs patterned with NSEs, accepting grammatical sentences and rejecting ungrammatical ones.

In summary, NSEs and intermediate JLEs correctly distinguished between grammatical and ungrammatical sentences regardless of verb type. In contrast, elementary JLEs incorrectly accepted ungrammatical unaccusative constructions to the same extent as grammatical ones but distinguished between verb types within ungrammatical conditions (i.e., ungrammatical unaccusatives vs. ungrammatical unergatives).

6. Discussion

The results obtained are interesting from several perspectives. First, our findings showed that elementary JLEs accepted the ungrammatical *Wh-Aux-VP-DP* word order for unaccusative verb constructions to a similar extent as the grammatical counterpart, *Wh-Aux-DP-VP*, while they rated the ungrammatical configuration for unergative verb constructions significantly lower. This suggests that they recognize the underlying distinction between unergatives and unaccusatives. Assuming that elementary L2 learners are generally strongly influenced by their L1 (e.g., Schwartz & Sprouse, 1996), we can infer that sensitivity to the unergative-unaccusative distinction is attributed to the learners' L1. Indeed, the fact that our elementary JLEs permitted both subject-in-situ and subject-ex-situ configurations suggests that subject displacement is optional, as it is in Japanese (i.e., via scrambling). Thus, assuming L1 transfer, the elementary learners' sensitivity to the unergative-unaccusative distinction may imply that this distinction is also present in their L1, Japanese. If this argument is correct, our data from L2 participants provide new (indirect) evidence for unaccusativity in Japanese.

In contrast, if unaccusativity is absent in Japanese, the source of our learners' judgments does not lie in their L1. In this case, a plausible and reliable source is UG (e.g., Baker, 1988). Our results showed that elementary learners were sensitive to the unergative-unaccusative distinction. If the source of this knowledge is UG, the results suggest that L2 learners are constrained by UG, contrary to arguments made by Meisel (1991, 1997), among others.

Finally, let us consider a remaining question regarding intermediate JLEs' grammar: Have the learners acquired $u\varphi$-driven subject movement? It is evident that they have successfully unlearned optional scrambling in their L2 English. However, it remains possible that they merely apply scrambling obligatorily, rather than acquiring $u\varphi$-driven movement. To uncover the nature of subject movement in their interlanguage, detailed examinations are imperative. Nevertheless, we would like to offer some notes here. Kimura (2022a, b) and Kimura and Wakabayashi (2024), proposing the *UG-as-Guide* (UGG) model, argue that the adoption of a 'wild' rule of forced scrambling triggers the activation of UG, reshaping the interlanguage to align with UG. Therefore, even if our learners employ a non-target-like forced scrambling rule, it is predicted that UG intervenes to facilitate the acquisition of the target-like grammar (or feature, namely, $u\varphi$). Given this, it would be particularly interesting, in light of UGG or UG-access theories, to investigate whether L2 learners can ultimately acquire the target $u\varphi$-feature.

7. Conclusion

We constructed erroneous sentences containing unaccusative verbs that elementary JLEs were likely to accept, based on previous insights. As a result of the acceptability judgment task, it was observed that they accepted both *DP-VP* and *VP-DP* orders for unaccusatives, whereas they only accepted the correct *DP-VP* order for unergatives. This phenomenon suggests two potential implications: (i) the learners' judgments are influenced by their L1, which provides new evidence for unaccusativity in Japanese; and (ii) unaccusativity may be absent in Japanese, but UG guides the learners toward the unergative-unaccusative distinction. Although further investigations are required to draw a solid conclusion, this line of research offers a promising avenue for future studies. The first implication contributes to theoretical discussions on Japanese syntax, while the second supports the UG-access hypothesis in late L2 acquisition.

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